Networking Principles, Fundamentals and Common Hardware

Part 1: Networking Principles

What is a network?

A network is **two or more devices connected to share data and resources** (files, printers, internet access).

Benefits

- Share files, printers and internet.
- Central management and backup.
- Communication (email, chat, video calls).
- Lower costs by sharing resources.

Types of networks (by size)

Type	Meaning	Example
PAN	Personal Area Network	Bluetooth between phone and earbuds
LAN	Local Area Network	Home, school or office network
WLAN	Wireless LAN	Wi-Fi network
CAN	Campus Area Network	University campus
MAN	Metropolitan Area Network	City-wide network
WAN	Wide Area Network	The **internet**
SAN	Storage Area Network	High-speed network for storage in data centres

Network models

- **Peer-to-peer (P2P):** every computer is equal and shares directly. Simple and cheap, suited to small networks, but harder to manage and less secure.
- **Client-server:** a central **server** provides services (files, authentication, web) to **clients**. Better security and control, but needs a server and administration.
- **Hybrid** models also exist.
- **Domain vs workgroup (Windows):** a **workgroup** is peer-to-peer; a **domain** uses a central server (Active Directory).

Network topologies

Topology	Description	Pros	Cons
Bus	All devices on one backbone cable	Cheap, simple	One break stops everything; slow with many devices
Star	All devices connect to a central switch	Easy to manage; one failure doesn't stop others	Central device is a single point of failure
Ring	Devices form a loop	Orderly data flow	One break can disrupt the ring
Mesh	Devices connect to many others	Very reliable	Expensive and complex
Hybrid	Mix of topologies	Flexible	Complex
Point-to-point	Two devices directly linked	Simple	Only two devices

Star is by far the most common today (Ethernet with a switch).

Measuring performance

- **Bandwidth:** maximum capacity, in bits per second (**bps**, **Kbps**, **Mbps**, **Gbps**).
- **Throughput:** the actual data rate achieved.
- **Latency:** delay, in milliseconds (ms).
- **Jitter:** variation in delay (affects calls and games).
- **Packet loss:** packets that never arrive.
- **Note:** network speed is in **bits**, while file sizes are in **bytes**. 1 byte = 8 bits, so 100 Mbps is about 12.5 MB/s at best.

Part 2: Networking Fundamentals

Data units

- Data is split into **packets** (and **frames** at the data-link layer), sent across the network, and reassembled at the destination.
- A **protocol** is a set of rules for communication, such as HTTP, TCP, IP and DNS.

Addresses

Address	Layer	Example	Purpose
MAC address	Data link (Layer 2)	00:1A:2B:3C:4D:5E	Unique hardware address of a NIC, 48 bits, written in hex

IP address	Network (Layer 3)	192.168.1.10	Logical address for routing between networks
Port number	Transport (Layer 4)	80, 443	Identifies the service or application

Communication types

- **Unicast:** one sender to one receiver.
- **Broadcast:** one sender to everyone on the network.
- **Multicast:** one sender to a chosen group.

Transmission modes

- **Simplex:** one direction only (a TV broadcast).
- **Half-duplex:** both directions but one at a time (walkie-talkie, old hubs).
- **Full-duplex:** both directions at the same time (modern switches and Ethernet).

Ethernet basics

- **Ethernet (IEEE 802.3)** is the standard wired LAN technology.
- Speeds: Fast Ethernet 100 Mbps, **Gigabit 1 Gbps**, 10 Gigabit and above.
- **CSMA/CD** was used in old half-duplex Ethernet to handle collisions. Modern full-duplex switches don't have collisions.
- **Wi-Fi** uses **CSMA/CA** (collision avoidance).

The OSI model

A 7-layer reference model for how networks communicate. A popular memory phrase is **"Please Do Not Throw Sausage Pizza Away"** (Physical up to Application).

Layer	Name	Job	Examples
7	**Application**	Services for apps	HTTP, FTP, SMTP, DNS
6	**Presentation**	Format, encryption, compression	SSL/TLS, JPEG
5	**Session**	Opens, manages and closes sessions	NetBIOS, RPC
4	**Transport**	End-to-end delivery, ports	TCP, UDP
3	**Network**	Addressing and routing	IP, routers
2	**Data link**	Frames, MAC addresses	Ethernet, switches
1	**Physical**	Bits over the medium	Cables, hubs, radio

- Data is **encapsulated** going down the layers (data, segment, packet, frame, bits) and **de-encapsulated** going up.

- The **TCP/IP model** is the practical version with four layers: **Application, Transport, Internet, Network access** (Day 16).

TCP vs UDP (preview)

- **TCP:** reliable, connection-oriented, checks delivery (web, email, file transfer).
- **UDP:** fast, connectionless, no guarantee (streaming, voice, games, DNS).

Part 3: Common Network Hardware

Device	OSI layer	Function
NIC (Network Interface Card)	1-2	Connects a device to the network (wired or Wi-Fi)
Hub	1	Repeats incoming signals to **all** ports; half-duplex; obsolete
Repeater / extender	1	Boosts a weak signal to extend distance
Bridge	2	Joins two network segments, filters by MAC
Switch	2	Sends frames only to the right port using the MAC address table
Router	3	Connects different networks and forwards packets by IP address
Layer 3 switch	2-3	Switch with routing ability
Access point (AP)	1-2	Provides Wi-Fi connection to a wired network
Modem	1	Converts signals between your network and the ISP line (cable, DSL, fibre ONT)
Firewall	3-7	Filters traffic by rules to protect the network
Gateway	Various	Connects networks that use different protocols; the default exit to other networks
Patch panel	1	Organises cable connections in a rack
PoE injector / switch	1	Delivers **power over Ethernet** to devices such as APs, IP cameras and VoIP phones
Load balancer	4-7	Spreads traffic across servers
NAS	n/a	Network storage device
VoIP phone, IP camera, printers	n/a	Network-connected endpoint devices

Hub vs switch vs router

	Hub	**Switch**	**Router**
Works with	Bits	MAC addresses	IP addresses
Sends data to	Every port	The correct port only	The correct network
Collisions	Yes, one shared domain	Each port is its own collision domain	Separates broadcast domains
Typical use	Obsolete	Inside a LAN	Between networks, to the internet

A **home router** usually combines router, switch, Wi-Fi access point and firewall in one box, with a separate **modem** or built-in modem.

Network cabling overview

- **Twisted pair (Cat 5e, 6, 6a):** RJ-45 Ethernet, up to 100 m per segment.
- **Fibre optic:** long distance and very high speed, immune to interference.
- **Coaxial:** cable internet and TV.
- **Wireless:** Wi-Fi, Bluetooth, cellular.
  (Days 15 and 17 cover cables and wireless in detail.)

Basic network commands

Command	**Use**
ipconfig (Windows), ip a or ifconfig (Linux)	Show IP configuration
ipconfig /all	Detailed information, including the MAC address, DHCP and DNS
ping 8.8.8.8	Test connectivity and delay
tracert (Windows), traceroute (Linux)	Show the path to a destination
nslookup	Test DNS
netstat	Show connections and ports
arp -a	Show the IP to MAC table

Exam points (Day 14)

- **LAN** is local, **WAN** is wide (the internet), **PAN** is personal (Bluetooth), **MAN** is city-wide.
- **Peer-to-peer** has no central server; **client-server** has one.
- **Star topology** is the most common and uses a central switch.
- A **MAC address** is a 48-bit hardware address (Layer 2); an **IP address** is a logical address (Layer 3).
- The **OSI model** has 7 layers: Physical, Data link, Network, Transport, Session, Presentation, Application.
- A **hub** repeats to all ports, a **switch** forwards by MAC address, and a **router** connects networks using IP addresses.
- An **access point** gives wireless access; a **modem** connects to the ISP.
- **Bandwidth** is capacity; **latency** is delay; network speeds are in bits per second.
- **Full-duplex** means sending and receiving at the same time.
- A **169.254.x.x** address means DHCP failed.
- ping tests connectivity, ipconfig shows IP settings, tracert shows the route.
- **PoE** carries power over the network cable.